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United States Patent Application Publication

20250283457

Kind Code

A1

Publication Date

September 11, 2025

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FLUID CONTROL DEVICE

Abstract

A fluid control device includes: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction. The pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole. The valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing.

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Family ID: 1000008528577

Appl. No.: 19/071037

Filed: March 05, 2025

Foreign Application Priority Data

JP 2024-032865

Mar. 05, 2024

Publication Classification

Int. Cl.: F04B39/10 (20060101); F04B45/047 (20060101)

U.S. Cl.:

CPC F04B39/10 (20130101); F04B45/047 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Japanese Patent Application No. 2024-032865 filed on Mar. 5, 2024. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] The present disclosure relates to a fluid control device configured to convey a fluid such as gas.

2. Description of the Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2016-200067 describes a fluid control device. The fluid control device described in Japanese Unexamined Patent Application Publication No. 2016-200067 includes a pump housing, a rapid discharge valve, and a heat sink. The rapid discharge valve includes an ejection hole and a discharge passage.

[0004] The heat sink is physically connected to the pump housing. Part of the heat sink is disposed at a position facing the discharge passage.

[0005] When driving a pump, the fluid control device supplies air to a cuff communicating with the ejection hole of the rapid discharge valve. When stopping driving the pump, the fluid control device discharges the air collected in the cuff from the discharge passage of the rapid discharge valve.

[0006] The air discharged from the discharge passage comes in contact with the heat sink and cools the heat sink, thus cooling the pump housing.

BRIEF SUMMARY OF THE DISCLOSURE

[0007] However, in such an existing fluid control device described in Japanese Unexamined Patent Application Publication No. 2016-200067, cooling of a pump housing requires a heat sink. Thus, the fluid control device is increased in size. In addition, the pump housing is indirectly cooled. Thus, it is difficult for the fluid control device to achieve high cooling efficiency.

[0008] Accordingly, a possible benefit of the present disclosure is to provide a small fluid control device having high cooling efficiency.

[0009] A fluid control device according to an embodiment of the present disclosure includes: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction. The pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole. The valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing.

[0010] The second valve housing member of the valve housing includes a projection formed by a part not overlapping an external shape of the pump housing when viewed in the first direction. The projection has at least part of the discharge passage. An outside end surface of the discharge passage opened to an outside faces the pump in the first direction. The vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction. At least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

[0011] With this configuration, the gas discharged from the discharge passage of the valve flows along an outer side surface of the pump housing. That is, a discharge flow along the outer side

surface of the pump housing is generated. Thus, the pump housing is directly cooled by the discharge flow without a heat sink, for example. In addition, the discharged gas flows also in the vicinity of the vibrator projection. Thus, the vibrator projection is directly cooled by the discharge flow, thus cooling the vibrator in the pump housing.

[0012] According to the present disclosure, it is possible to form a small fluid control device having high cooling efficiency.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0013] FIG. 1 is an external perspective view of a fluid control device according to an embodiment of the present disclosure;

[0014] FIG. 2 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure;

[0015] FIG. 3 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure;

[0016] FIG. 4 is a plan view of the fluid control device according to the embodiment of the present disclosure when viewed in a first direction; and

[0017] FIG. 5 is a side sectional view illustrating the configuration of the fluid control device according to the embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0018] A fluid control device according to an embodiment of the present disclosure will be described with reference to the drawings. FIG. 1 is an external perspective view of a fluid control device according to an embodiment of the present disclosure. Each of FIGS. 2 and 3 is an exploded perspective view of the fluid control device according to the embodiment of the present disclosure. FIG. 4 is a plan view of the fluid control device according to the embodiment of the present disclosure when viewed in a first direction. FIG. 5 is a side sectional view illustrating the configuration of the fluid control device according to the embodiment of the present disclosure. FIG. 5 is a schematic side sectional view for facilitating understanding of the configuration of the fluid control device.

[0019] A fluid control device **10** includes a vibrator **20**, a plurality of housing members **31** to **37**, a membrane **40**, a membrane **50**, and a membrane fixing member **400**. The housing members **34** and **35** correspond to housing members forming a pump housing and correspond to part of a first valve housing member. The housing member **36** corresponds to part of the first valve housing member. The housing member **37** corresponds to a second valve housing member. In the fluid control device **10**, an aspect in which housing members common to the pump housing and a valve housing are used is illustrated. However, housing members forming the pump housing and housing members forming the valve housing may be separate from each other.

[0020] The vibrator **20** includes a plurality of flat plates **211**, **212**, and **213** and a plurality of piezoelectric bodies **221** and **222**. The plurality of flat plates **211**, **212**, and **213** are made of a metal and are configured to be able to deform (vibrate) by being subjected to a stress generated by distortion of the plurality of piezoelectric bodies **221** and **222**.

[0021] The flat plate **211** has an outer peripheral end **E211** having a substantially circular shape in plan view. The flat plate **211** includes a projection **P211** projecting outward from the outer peripheral end **E211**. For example, the projection **P211** has a substantially rectangular shape in plan view.

[0022] The flat plate **212** has an outer peripheral end **E212** having a substantially circular shape in plan view. The flat plate **212** includes a projection **P212** projecting outward from the outer peripheral end **E212**. For example, the projection **P212** has a substantially rectangular shape in plan

view.

[0023] The flat plate **213** has an outer peripheral end **E213** having a substantially circular shape in plan view. The flat plate **213** includes projections **P213** projecting outward from the outer peripheral end **E213**. For example, the projections **P213** have substantially rectangular shapes in plan view.

[0024] The outer peripheral end **E211** of the flat plate **211**, the outer peripheral end **E212** of the flat plate **212**, and the outer peripheral end **E213** of the flat plate **213** have substantially the same shape. The projection **P211**, the projection **P212**, and the projections **P213** each correspond to a vibrator projection of the present disclosure.

[0025] The plurality of piezoelectric bodies **221** and **222** have a circular shape in plan view.

[0026] In the vibrator **20**, the flat plate **211**, the piezoelectric body **221**, the flat plate **212**, the piezoelectric body **222**, and the flat plate **213** are laminated in this order in the first direction. The flat plate **211**, the piezoelectric body **221**, the flat plate **212**, the piezoelectric body **222**, and the flat plate **213** are laminated such that the respective centers of the flat plate **211**, the piezoelectric body **221**, the flat plate **212**, the piezoelectric body **222**, and the flat plate **213** substantially coincide with each other when viewed in the first direction. Thus, the vibrator **20** forms a bimorph actuator.

[0027] In this case, the outer peripheral ends **E211**, **E212**, and **E213** of the plurality of flat plates **211**, **212**, and **213** substantially coincide with each other when viewed in the first direction. Then, the position where the outer peripheral ends **E211**, **E212**, and **E213** overlap each other corresponds to an outer peripheral end **E20** of the vibrator **20**.

[0028] The housing member **31** is a flat plate having an outer peripheral end **E31** having a substantially circular shape in plan view. The outer peripheral end **E31** and the outer peripheral end **E20** of the vibrator **20** have substantially the same shape. The housing member **31** includes a projection **P31** projecting outward from the outer peripheral end **E31**.

[0029] The housing member **31** has a through hole **311** and a plurality of through holes **312**. The through hole **311** and the plurality of through holes **312** have circular shapes when viewed from the side where a surface where each through hole is open is located. The through hole **311** is formed at a position including the center of the housing member **31** in plan view. The center of the through hole **311** substantially coincides with the center of a circle formed by the outer peripheral end **E31**. The plurality of through holes **312** are formed between the through hole **311** and the outer peripheral end **E31** at intervals of a substantially equal angle in a circumferential direction surrounding the through hole **311**.

[0030] The housing member **32** is a circular ring-shaped body having an outer peripheral end **E32** having a substantially circular shape in plan view. The outer peripheral end **E32** and the outer peripheral end **E31** have substantially the same shape. The housing member **32** includes projections **P32** projecting outward from the outer peripheral end **E32**.

[0031] The housing member **32** is a ring-shaped body having a through hole **321**. The diameter of the through hole **321** is larger than the diameter of a circle formed by the plurality of through holes **312** of the housing member **31**.

[0032] The housing member **33** is a circular ring-shaped body having an outer peripheral end **E33** having a substantially circular shape in plan view. The outer peripheral end **E33** and the outer peripheral ends **E31** and **E32** have substantially the same shape. The housing member **33** includes projections **P33** projecting outward from the outer peripheral end **E33**.

[0033] The housing member **33** has a through hole **331** as a space inside the ring-shaped body. The diameter of the through hole **331** is substantially equal to the diameter of the through hole **321** of the housing member **32**.

[0034] The housing member **34** is a flat plate having an outer peripheral end **E34** having a substantially circular shape in plan view. The outer peripheral end **E34** and the outer peripheral end **E33** have substantially the same shape. The housing member **34** includes projections **P34** projecting outward from the outer peripheral end **E34**.

[0035] The housing member **34** has a through hole **341** and a plurality of through holes **342**. The through hole **341** and the plurality of through holes **342** have circular shapes when viewed from the side where a surface where each through hole is open is located. The through hole **341** is formed at a position including the center of the housing member **34** in plan view. The center of the through hole **341** substantially coincides with the center of a circle formed by the outer peripheral end **E34**. The plurality of through holes **342** are circumferentially formed between the through hole **341** and the outer peripheral end **E34**.

[0036] The housing member **35** is a flat plate having an outer peripheral end **E35** having a substantially circular shape in plan view. The outer peripheral end **E35** and the outer peripheral end **E34** have substantially the same shape. The housing member **35** includes projections **P35** projecting outward from the outer peripheral end **E35**.

[0037] The housing member **35** has a recess **351**, a plurality of grooves **352**, and a plurality of through holes **353**. The recess **351** has a shape recessed from one main surface of the housing member **35**. The recess **351** has a circular shape when viewed from the side where a surface where the recess **351** is open is located. The recess **351** is formed at a position including the center of the housing member **35** in plan view. The center of the recess **351** substantially coincides with the center of a circle formed by the outer peripheral end **E35**.

[0038] As with the recess **351**, the plurality of grooves **352** each have a shape recessed from the one main surface of the housing member **35**. The plurality of grooves **352** each have a belt shape. The plurality of grooves **352** are formed radially from the recess **351** to the outer peripheral end **E35**. The plurality of grooves **352** are formed at intervals of a predetermined angle in a circumferential direction surrounding the recess **351**. The plurality of grooves **352** reach the outer peripheral end **E35** and are open to the outside from the outer peripheral end **E35**.

[0039] The plurality of through holes **353** have a circular shape when viewed from the side where a surface where each through hole is open is located. The plurality of through holes **353** are circumferentially formed between the recess **351** and the outer peripheral end **E35**. The plurality of through holes **353** are formed between the plurality of grooves **352** in the circumferential direction surrounding the recess **351**.

[0040] The housing member **36** is a flat plate having an outer peripheral end **E36** having a substantially circular shape in plan view. The outer peripheral end **E36** and the outer peripheral end **E35** have substantially the same shape. The housing member **36** includes projections **P36** projecting outward from the outer peripheral end **E36**.

[0041] The housing member **36** includes a recess **361**, a plurality of through holes **362**, and a seat **360**. The recess **361** has a shape recessed from one main surface of the housing member **36**. The recess **361** has a circular shape when viewed from the side where a surface where the recess **361** is open is located. The recess **361** is formed at a position including the center of the housing member **36** in plan view. The center of the recess **361** substantially coincides with the center of a circle formed by the outer peripheral end **E36**.

[0042] The plurality of through holes **362** each have an elliptical shape when viewed from the side where a surface where each through hole is open is located. The plurality of through holes **362** are circumferentially formed around the center of the housing member **36**. The plurality of through holes **362** are each formed such that the long-side direction of the elliptical shape is parallel to a direction extending radially from the center of the housing member **36**. The plurality of through holes **362** overlap the recess **361** in plan view and communicate with a recess internal space formed by the recess **361** and the membrane **50**.

[0043] The seat **360** has a cylindrical shape. The seat **360** has a shape projecting from a bottom surface of the recess **361**. The seat **360** is formed at a position including the center of the housing member **36** in plan view.

[0044] The housing member **37** includes a housing member **371** and a housing member **372**.

[0045] The housing member **371** is a flat plate having an outer peripheral end **E371** having a

substantially circular shape in plan view. The shape of the outer peripheral end E371 is larger than the shape of each of the outer peripheral ends E31 to E36. The housing member 371 includes projections projecting outward from the outer peripheral end E371.

[0046] The housing member 371 has a through hole 3711, through holes 3712, grooves 3713, and through holes 3714.

[0047] The through hole 3711 has a circular shape when viewed from the side where a surface where the through hole is open is located. The through hole 3711 is formed at a position including the center of the housing member 371 in plan view. The center of the through hole 3711 substantially coincides with the center of a circle formed by the outer peripheral end E371.

[0048] The through holes 3712 and 3714 have a circular shape when viewed from the side where the surface where each through hole is open is located. The diameter of the through holes 3712 and 3714 is smaller than the diameter of the through hole 3711. The through holes 3712 and 3714 are formed between the through hole 3711 and the outer peripheral end E371. The through holes 3712 and 3714 are arranged in directions extending radially from the center of the housing member 37. The through holes 3714 are formed at positions closer to the outer peripheral end E371 than the through holes 3712.

[0049] The grooves 3713 each have a shape recessed from one main surface of the housing member 371. The groove 3713 is formed between the through hole 3712 and the through hole 3714 in plan view. The groove 3713 communicates with the through hole 3712 and the through hole 3714.

[0050] Two pairs of the through holes 3712, the grooves 3713, and the through holes 3714 are formed. The pairs are disposed at positions located point-symmetrically relative to the center of the housing member 37.

[0051] The housing member 372 is a flat plate having an outer peripheral end E372 having a substantially circular shape in plan view. The outer peripheral end E372 and the outer peripheral end E371 have the same shape. The housing member 372 includes projections projecting outward from the outer peripheral end E372.

[0052] The housing member 372 has a through hole 3721. The through hole 3721 has a circular shape when viewed from the side where a surface where the through hole is open is located. The through hole 3721 is formed at a position including the center of the housing member 372 in plan view. The center of the through hole 3721 substantially coincides with the center of a circle formed by the outer peripheral end E372. The diameter of the through hole 3721 is smaller than the diameter of the through hole 3711 of the housing member 371.

[0053] The housing member 372 is in contact with a surface of the housing member 371 where the grooves 3713 are open. Thus, the housing member 371 and the housing member 372 have discharge passages formed by the through holes 3712, the grooves 3713 whose openings are closed by the housing member 371, and the through holes 3714. The housing member 371 and the housing member 372 may be formed separately from each other or integrally with each other.

[0054] The membrane 40 is made of a flexible material. The membrane 40 has a circular shape.

[0055] The membrane 50 is made of a flexible material. The membrane 50 has a circular shape. The membrane 50 has a through hole 500. The through hole 500 has a circular shape when viewed from the side where a surface where the through hole is open is located. The diameter of the through hole 500 is smaller than the diameter of the seat 360 of the housing member 36.

[0056] The housing member 31 and the housing member 32 are disposed to be laminated on one side of the vibrator 20 in the first direction. The housing members 33 to 37 are disposed to be laminated on the other side of the vibrator 20 in the first direction. More specifically, this configuration is as follows.

[0057] The housing member 32 is disposed on the side of the vibrator 20 closer to the flat plate 211 and is connected to the flat plate 211. In this case, the outer peripheral end E32 of the housing member 32 coincides with the outer peripheral end E20 of the vibrator 20 when viewed in the first

direction.

[0058] The housing member **31** is disposed on the opposite side of the housing member **32** from the vibrator **20** and is connected to the housing member **32**. In this case, the outer peripheral end **E31** of the housing member **31** coincides with the outer peripheral end **E32** of the housing member **32** when viewed in the first direction.

[0059] The housing member **33** is disposed on the side of the vibrator **20** closer to the flat plate **213** and is connected to the flat plate **213**. In this case, the outer peripheral end **E33** of the housing member **33** coincides with the outer peripheral end **E20** of the vibrator **20** when viewed in the first direction.

[0060] The housing member **34** is disposed on the opposite side of the housing member **33** from the vibrator **20** and is connected to the housing member **33**. In this case, the outer peripheral end **E34** of the housing member **34** coincides with the outer peripheral end **E33** of the housing member **33** when viewed in the first direction.

[0061] The through hole **341** and the plurality of through holes **342** of the housing member **34** communicate with the through hole **331** of the housing member **33**.

[0062] The housing member **35** is disposed on the opposite side of the housing member **34** from the housing member **33** and is connected to the housing member **34**. In this case, the housing member **35** is connected to the housing member **34** such that the side of the housing member **35** where the recess **351** and the plurality of grooves **352** are open is closer to the housing member **34**. In addition, the outer peripheral end **E35** of the housing member **35** coincides with the outer peripheral end **E34** of the housing member **34** when viewed in the first direction.

[0063] The recess **351** and the plurality of grooves **352** of the housing member **35** communicate with the through hole **341** of the housing member **34**. The plurality of through holes **353** of the housing member **35** communicate with the plurality of respective through holes **342** of the housing member **34**.

[0064] The housing member **36** is disposed on the opposite side of the housing member **35** from the housing member **34** and is connected to the housing member **35**. In this case, the housing member **36** is connected to the housing member **35** such that the opposite side of the housing member **36** from the side where the recess **361** is open is closer to the housing member **35**. In addition, the outer peripheral end **E36** of the housing member **36** coincides with the outer peripheral end **E35** of the housing member **35** when viewed in the first direction.

[0065] The plurality of through holes **362** of the housing member **36** communicate with the plurality of respective through holes **353** of the housing member **35**.

[0066] The housing member **371** is disposed on the opposite side of the housing member **36** from the housing member **35** and is connected to the housing member **36**. In this case, the outer peripheral end **E371** of the housing member **371** forming the housing member **37** and the outer peripheral end **E372** of the housing member **372** forming the housing member **37** coincide with the outer peripheral end **E36** of the housing member **36** when viewed in the first direction.

[0067] The through hole **3711** and the through holes **3712** of the housing member **37** communicate with the recess **361** of the housing member **36**. The through holes **3714** of the housing member **37** are located outside the outer peripheral end **E36** of the housing member **36**. In the housing member **37**, the through hole **3711** and the through hole **3721** communicate with each other.

[0068] The membrane **40** is disposed in a space surrounded by the vibrator **20**, the housing member **33**, and the housing member **34**. The membrane **40** is fixed with the membrane fixing member **400** to cover the central position of the flat plate **213**. A central portion of the membrane **40** is fixed, and a peripheral portion of the membrane **40** is movably fixed.

[0069] The membrane **50** is disposed in a space (the recess **361**) surrounded by the housing member **36** and the housing member **37**. The membrane **50** is fixed by holding an outer peripheral portion of the membrane **50** between the housing member **36** and the housing member **37**. A central portion of the membrane **50** is movably fixed. The through hole **500** of the membrane **50** overlaps

the seat **360** when viewed in the first direction.

[0070] With this configuration, when a driving voltage is applied to the vibrator **20**, the vibrator **20** generates bending vibrations. The application of a driving voltage to the vibrator **20** is performed by using the flat plates **211**, **212**, and **213**, for example. In this case, it is possible to use each projection.

[0071] Bending vibrations of the vibrator **20** vary the volume of the space (pump space) surrounded by the vibrator **20**, the housing member **33**, and the housing member **34**. Thus, the gas outside the fluid control device **10** is suctioned into the space (pump space) surrounded by the vibrator **20**, the housing member **33**, and the housing member **34** from the grooves **352** through the recess **351** and the through hole **341**.

[0072] The gas suctioned into the space surrounded by the vibrator **20**, the housing member **33**, and the housing member **34** flows into the space (valve space) surrounded by the housing member **36** and the housing member **37** through the through holes **342**, the through holes **353**, and the through holes **362**.

[0073] In this manner, the fluid control device **10** includes the pump housing formed by the plurality of housing members **31** to **36**, and the vibrator **20** and thus functions as a pump. Then, the grooves **352** each correspond to a suction hole (first hole) of the fluid control device **10**, and the through holes **342**, the through holes **353**, and the through holes **362** each correspond to a communication hole (second hole) of the fluid control device **10**.

[0074] In this case, the membrane **40** is deformed by a gas flow. Thus, the membrane **40** does not close a passage when gas is suctioned into the pump space from the grooves **352** and closes the passage when gas flows into the pump space from the through holes **342**. Accordingly, the fluid control device **10** can reduce the backflow of gas.

[0075] The gas that has flowed into the space (valve space) surrounded by the housing member **36** and the housing member **37** pushes up the membrane **50**, and the membrane **50** is then separated from a seat **360**. Thus, the gas flows into the through hole **3711** and the through hole **3721** of the housing member **37** through the through hole **500** of the membrane **50** and is ejected to the outside of the fluid control device **10** from the through hole **3721**. That is, the through hole **3721** corresponds to an ejection hole (third hole) of the fluid control device **10**.

[0076] In this case, the membrane **50** closes the through holes **3712**. Thus, the fluid control device **10** can inhibit the leakage of gas through the through holes **3712**, the grooves **3713**, and the through holes **3714**.

[0077] When the application of a driving voltage to the vibrator **20** is stopped, bending vibrations of the vibrator **20** are stopped. In this case, for example, when a cuff (not illustrated) is set to the through hole **3721**, gas flows backward from the cuff through the through hole **3721**. The backflow of the gas pushes down the membrane **50**, and the membrane **50** comes in contact with the seat **360**. Thus, the through hole **500** of the membrane **50** is closed by the seat **360**. Accordingly, the fluid control device **10** can inhibit the back-flown gas from flowing backward toward the pump.

[0078] Then, the movement of the membrane **50** toward the seat **360** opens the through holes **3712**. Thus, the back-flown gas is discharged through the through holes **3712**, the grooves **3713**, and the through holes **3714**. That is, the through holes **3712**, the grooves **3713**, and the through holes **3714** correspond to the discharge passages of the fluid control device **10**.

[0079] In this manner, the fluid control device **10** includes the valve housing formed by the plurality of housing members **36** and **37**, and the membrane **50** and thus functions as a valve.

[0080] In this configuration, the shape of an outer peripheral end E37 of the housing member **37** is larger than the shape of each of the outer peripheral ends E31 to E36 of the other housing members **31** to **36** forming the fluid control device **10**. Thus, the through holes **3714** forming the discharge passages are located outside an outer peripheral end of the pump housing.

[0081] In addition, the through holes **3714** are open toward the housing members **31** to **36**. That is, the end surfaces (outside end surfaces) OP3712 of the discharge passages exposed to the outside

are exposed to face the pump housing.

[0082] Thus, the gas discharged through each discharge passage (discharge flow) flows toward the pump housing (see discharge flows represented by thick arrows in FIGS. 1 and 5). The gas discharged from each discharge passage flows along an outer side surface (outer surface formed by the outer peripheral ends E31 to E36 of the plurality of housing members 31 to 36) of the pump housing. That is, a discharge flow along the outer side surface of the pump housing is generated. Thus, the pump housing is directly cooled by the discharge flow without a heat sink, for example. Accordingly, the fluid control device 10 can achieve high cooling efficiency with a small size.

[0083] In addition, when viewed in the first direction, the plurality of projections P211, P212, and P213 of the vibrator 20 do not overlap the end surfaces OP3712 of the discharge passages exposed to the outside. Thus, part of the discharged gas is not blocked by the pump housing and the plurality of projections P211, P212, and P213 of the vibrator 20, and the gas flow is not obstructed. In addition, it is possible to inhibit the discharge flow from being disturbed by the plurality of projections P211, P212, and P213. Accordingly, the fluid control device 10 can achieve high cooling efficiency.

[0084] When viewed in the first direction, the projections of the plurality of housing members 31 to 36 also do not overlap the end surfaces OP3712 of the discharge passages exposed to the outside. Thus, it is possible to inhibit the discharge flow from being blocked and disturbed by the projections of the plurality of housing members 31 to 36. Accordingly, the fluid control device 10 can achieve high cooling efficiency.

[0085] In addition, when viewed in the first direction, the end surfaces OP3712 of the discharge passages exposed to the outside may be as close to the outer peripheral end of the pump housing (outer peripheral ends E31 to E36 of the plurality of housing members 31 to 36) as possible. Furthermore, when viewed in the first direction, the end surfaces OP3712 of the discharge passages exposed to the outside may have parts overlapping the outer peripheral end of the pump housing. Thus, the effect of cooling by the discharged gas is further improved.

[0086] In addition, the number of the discharge passages may be two or more. This increases the number of parts of the pump housing to be cooled, thus improving the cooling effect. For example, in the present embodiment, two discharge passages are used to further improve the cooling effect than that in the case of having one discharge passage. Furthermore, three or more discharge passages can improve the cooling effect.

[0087] In addition, the plurality of discharge passages may be disposed at positions located point-symmetrically relative to the center of the valve housing when viewed in the first direction. This reduces variations in parts to be cooled in a circumferential direction, thus improving the cooling effect.

[0088] In addition, in the fluid control device 10, the plurality of flat plates 211, 212, and 213 respectively include the projections P211, P212, and P213. The projections P211, P212, and P213 project outward from the pump housing. Accordingly, the discharged gas flows also in the vicinity of each of the projections P211, P212, and P213. As a result, the projections P211, P212, and P213 are directly cooled by the discharge flows, thus cooling the vibrator 20 in the pump housing.

[0089] In this case, the flat plate 211 including the projection P211, the flat plate 212 including the projection P212, and the flat plate 213 including the projections P213 are made of a metal and thus have a high thermal conductivity. Accordingly, the fluid control device 10 can achieve a higher cooling effect.

[0090] <1> A fluid control device comprising: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction, wherein the pump has a first hole formed in the pump housing, a second hole formed in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to be able to send gas from the first hole to the second hole, the valve has the second hole formed in a first valve housing member of the valve housing, the first valve housing member being connected to the pump

housing, a third hole formed in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage formed in the second valve housing member of the valve housing, the second valve housing member includes a projection formed by a part not overlapping an external shape of the pump housing when viewed in the first direction, the projection has at least part of the discharge passage, an outside end surface of the discharge passage opened to an outside faces the pump in the first direction, the vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction, and at least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

[0091] <2> The fluid control device according to <1>, wherein the outside end surface of the discharge passage does not overlap the vibrator projection when viewed in the first direction.

[0092] <3> The fluid control device according to <1> or <2>, further comprising a plurality of the discharge passages, wherein the discharge passages are disposed at positions located point-symmetrically relative to a center of the valve housing when viewed in the first direction.

[0093] <4> The fluid control device according to any one of <1> to <3>, wherein the outside end surface of the discharge passage has a part overlapping the pump housing when viewed in the first direction.

[0094] <5> The fluid control device according to any one of <1> to <4>, wherein the vibrator has a multilayer structure including a piezoelectric body and a vibration body made of a metal, and the vibration body is held by the pump housing and includes the vibrator projection.

Claims

1. A fluid control device comprising: a pump including a pump housing; and a valve including a valve housing, the pump and the valve being disposed side by side in a first direction, wherein the pump has a first hole provided in the pump housing, a second hole provided in the pump housing, and a vibrator disposed in the pump housing, the vibrator being configured to send gas from the first hole to the second hole, the valve has the second hole provided in a first valve housing member of the valve housing, the first valve housing member being connected to the pump housing, a third hole provided in a second valve housing member of the valve housing, the second valve housing member being disposed to face the first valve housing member, and a discharge passage provided in the second valve housing member of the valve housing, the second valve housing member includes a projection provided by a part not overlapping an external shape of the pump housing when viewed in the first direction, the projection has at least part of the discharge passage, an outside end surface of the discharge passage opened to an outside faces the pump in the first direction, the vibrator includes a vibrator projection projecting from the pump housing when viewed in the first direction, and at least part of the outside end surface of the discharge passage is located outside the pump housing when viewed in the first direction.

2. The fluid control device according to claim 1, wherein the outside end surface of the discharge passage does not overlap the vibrator projection when viewed in the first direction.

3. The fluid control device according to claim 1, wherein the discharge passage includes a plurality of discharge passages, and wherein the plurality of discharge passages is disposed at positions located point-symmetrically relative to a center of the valve housing when viewed in the first direction.

4. The fluid control device according to claim 1, wherein the outside end surface of the discharge passage has a part overlapping the pump housing when viewed in the first direction.

5. The fluid control device according to claim 1, wherein the vibrator has a multilayer structure including a piezoelectric body, and a vibration body comprising a metal, and the vibration body is held by the pump housing and includes the vibrator projection.
